Vectors

Chapter Overview 2

Vector arithmetic

Add, subtract and scale vectors. Show that vectors are parallel by expressing one as a scalar multiple of the other.

Magnitude and direction

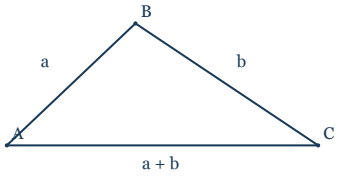
Calculate lengths, directions and unit vectors using components.

Position vectors and geometry

Describe points relative to an origin; solve problems involving ratios, parallel lines and intersections.

Modelling

Use vectors for displacement, bearings and forces. Distinguish speed from velocity.



Magnitude and direction

A vector describes a displacement. It has both magnitude (length) and direction.

$\overrightarrow{PQ}$ means the displacement from P to Q.

Vectors with the same magnitude and direction are equal, wherever they are drawn.

The triangle law

$$\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$$

Following one displacement and then another gives their resultant.

$$\overrightarrow{BA} = -\overrightarrow{AB}$$

Reversing the journey preserves the magnitude and reverses the direction.

What does multiplication mean?

The meaning of an operation depends on the mathematical objects involved. Vector addition is componentwise:

$$\begin{pmatrix} 3 \\ -2 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$$

The cross product of two three-dimensional vectors is perpendicular to both:

$$\begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \\ -2 \end{pmatrix}$$

$\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$

Cartesian products of sets

For $A = \{a, b, c\}$ and $B = \{d, e\}$, the Cartesian product consists of ordered pairs:

$$A \times B = \{(a, d), (a, e), (b, d), \\ (b, e), (c, d), (c, e)\}$$

$|A \times B| = |A|\,|B| = 3 \times 2 = 6$

Order matters: in general A $\times$ B and B $\times$ A are different sets. This is a Cartesian product, not a vector cross product.

Subtraction and the zero vector

$\mathbf{a} - \mathbf{b} = \mathbf{a} + (-\mathbf{b})$

$\overrightarrow{PQ} + \overrightarrow{QP} = \mathbf{0} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

The zero vector represents no displacement.

Scalar multiples

For a nonzero vector $\mathbf{a}$, multiplying by a positive scalar preserves direction; a negative scalar reverses direction.

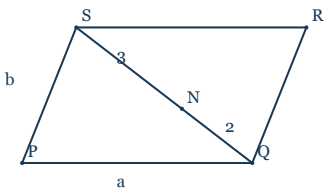
$|\lambda\mathbf{a}| = |\lambda|\,|\mathbf{a}|$

Nonzero scalar multiples are parallel.

Show that these vectors are parallel:

$2\mathbf{a} + 4\mathbf{b}, \qquad 3\mathbf{a} + 6\mathbf{b}$

Proof



PQRS is a parallelogram. N lies on SQ with SN:NQ = 3:2. Given $\overrightarrow{PQ} = \mathbf{a}$ and $\overrightarrow{PS} = \mathbf{b}$, find the following vectors.

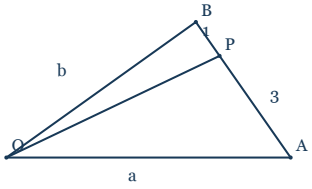
(a) $\overrightarrow{SQ}$

a

(b) $\overrightarrow{NR}$

b

Use the route N to Q to R. QR equals PS because opposite sides of a parallelogram have equal displacement vectors.



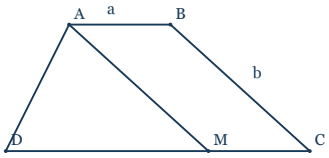
OAB is a triangle with $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$. P lies on AB with AP:PB = 3:1.

(a) Express $\overrightarrow{AB}$ in terms of a and b.

a

(b) Express $\overrightarrow{OP}$ in terms of a and b.

b



6. ABCD is a trapezium with AB parallel to DC and DC = 3AB. M lies on DC with DM:MC = 2:1.

$\overrightarrow{AB} = \mathbf{a}$, $\overrightarrow{BC} = \mathbf{b}$

Find AM, BD, MB and DA in terms of a and b.

6(a)

6(b)

6(c)

6(d)

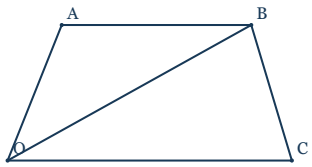
9. In triangle ABC, $\overrightarrow{AB} = \mathbf{a}$, $\overrightarrow{AC} = \mathbf{b}$. P and Q are the midpoints of AB and AC.

(a) Find BC, AP, AQ and PQ.

9(a)

(b) Show that PQ is parallel to BC.

9(b)



10. OABC is a quadrilateral with $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OC} = 3\mathbf{b}$, $\overrightarrow{OB} = \mathbf{a} + 2\mathbf{b}$.

(a) Find AB and CB in terms of a and b.

10(a)

(b) Show that AB is parallel to OC.

10(b)

11. The vectors 2a + kb and 5a + 3b are parallel. Find k, assuming a and b are nonparallel.

11

Representing Vectors 10

Components describe displacement

The vector $\binom{3}{-2}$ means 3 units right and 2 units down.

$$\mathbf{a} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

Find a + b and 2a.

a + b

2a

The unit vectors i and j

A unit vector has magnitude 1.

$$\mathbf{i} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \mathbf{j} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 4 \\ 3 \end{pmatrix} = 4\mathbf{i} + 3\mathbf{j}$$

If $P = (5, -1)$, then $\overrightarrow{OP} = 5\mathbf{i} - \mathbf{j}$.

A basis for the plane

Any two nonzero, nonparallel vectors can be used as a basis. Every vector in the plane can be expressed as a linear combination of them.

$$\begin{pmatrix} x \\ y \end{pmatrix} = p\begin{pmatrix} 5 \\ 2 \end{pmatrix} + q\begin{pmatrix} 3 \\ 0 \end{pmatrix}$$

$\mathbf{a} = 3\mathbf{i}, \quad \mathbf{b} = \mathbf{i} + \mathbf{j}, \quad \mathbf{c} = \mathbf{i} - 2\mathbf{j}$

1. Write a as a column vector.

1

2. Find b + 2c in i, j form.

2

Exercise 11B - Year 1 Pages 237-238 12

4

$\mathbf{a} = 2\mathbf{i} + 5\mathbf{j}, \quad \mathbf{b} = 3\mathbf{i} - \mathbf{j}$

(a) Find λ if $\mathbf{a} + \lambda\mathbf{b}$ is parallel to i.

4(a)

(b) Find μ if $\mu\mathbf{a} + \mathbf{b}$ is parallel to j.

4(b)

5

$\mathbf{c} = 3\mathbf{i} + 4\mathbf{j}, \quad \mathbf{d} = \mathbf{i} - 2\mathbf{j}$

(a) Find λ if $\mathbf{c} + \lambda\mathbf{d} \parallel \mathbf{i} + \mathbf{j}$.

5(a)

(b) Find μ if $\mu\mathbf{c} + \mathbf{d} \parallel \mathbf{i} + 3\mathbf{j}$.

5(b)

(c) Find s if $\mathbf{c} - s\mathbf{d} \parallel 2\mathbf{i} + \mathbf{j}$.

5(c)

(d) Find t if $\mathbf{d} - t\mathbf{c} \parallel -2\mathbf{i} + 3\mathbf{j}$.

5(d)

6

In triangle ABC, $\overrightarrow{AB} = 4\mathbf{i} + 3\mathbf{j}$ and $\overrightarrow{AC} = 5\mathbf{i} + 2\mathbf{j}$. Find $\overrightarrow{BC}$.

6

7

OABC is a parallelogram. P divides AC in the ratio AP:PC = 3:2. Given $\overrightarrow{OA} = 2\mathbf{i} + 4\mathbf{j}$ and $\overrightarrow{OC} = 7\mathbf{i}$, find each vector in i, j form and column form.

(a) $\overrightarrow{AC}$

7(a)

(b) $\overrightarrow{OP}$

7(b)

(c) $\overrightarrow{AP}$

7(c)

8

$\mathbf{a} = \begin{pmatrix} j \\ 3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 10 \\ k \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$

Given b − 2a = c, find j and k.

8

9

$\mathbf{a} = \begin{pmatrix} p \\ -q \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} q \\ p \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} 7 \\ 4 \end{pmatrix}$

Given a + 2b = c, find p and q.

9

10

The resultant of $\mathbf{a} = 3\mathbf{i} - 2\mathbf{j}$ and $\mathbf{b} = p\mathbf{i} - 2p\mathbf{j}$ is parallel to $2\mathbf{i} - 3\mathbf{j}$.

(a) Find p.

10(a)

(b) Find the resultant.

10(b)

Magnitude of a Vector 15

The magnitude of a vector is its length.

$$\left|\begin{pmatrix} x \\ y \end{pmatrix}\right| = \sqrt{x^2 + y^2}$$

Use Pythagoras on the horizontal and vertical components.

$\left|\begin{pmatrix} 3 \\ 4 \end{pmatrix}\right|$

1

$\left|\begin{pmatrix} 1 \\ -1 \end{pmatrix}\right|$

2

$\left|\begin{pmatrix} -5 \\ -12 \end{pmatrix}\right|$

3

$\left|\begin{pmatrix} 4 \\ -1 \end{pmatrix}\right|$

4

$\left|\begin{pmatrix} 2 \\ 0 \end{pmatrix}\right|$

5

Unit Vectors 16

A unit vector has magnitude 1.

To obtain a unit vector in the same direction as a nonzero vector, divide by its magnitude:

$$\hat{\mathbf{a}} = \frac{\mathbf{a}}{|\mathbf{a}|}$$

Example

$\mathbf{a} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$

Example

1. Find a unit vector in this direction.

$\begin{pmatrix} 12 \\ -5 \end{pmatrix}$

1

2. Find a unit vector in this direction.

$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

2

8

$|2\mathbf{i} - k\mathbf{j}| = 2\sqrt{10}$

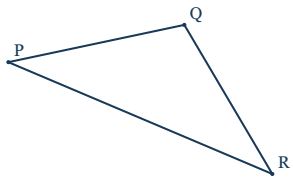
Find the exact values of k.

8

9

The vector $\mathbf{a} = p\mathbf{i} + q\mathbf{j}$ has magnitude 10 and makes an unsigned angle θ with the positive x-axis, where $\sin\theta = \frac{3}{5}$. Find the possible values of p and q.

9



11

In triangle PQR, $\overrightarrow{PQ} = 4\mathbf{i} + \mathbf{j}$ and $\overrightarrow{PR} = 6\mathbf{i} - 8\mathbf{j}$.

(a) Find angle QPR to 1 decimal place.

a

(b) Find the area of triangle PQR.

b

Use the unrounded angle in the area calculation.

Position Vectors 19

Position vectors

The position vector of a point A is $\overrightarrow{OA}$, where O is the origin.

$A = (3, 2) \iff \overrightarrow{OA} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$

Coordinates specify a point; a vector specifies a displacement. A position vector connects the two.

Translation example

Translate A by $\binom{4}{0}$ to obtain B.

B

Example 20

A and B have coordinates (3,4) and (11,2). Give each vector in i, j form.

(a) The position vector of A

a

(b) The position vector of B

b

(c) The vector $\overrightarrow{AB}$

c

Subtract the starting position vector from the finishing position vector.

$\overrightarrow{OA} = 5\mathbf{i} - 2\mathbf{j}, \qquad \overrightarrow{AB} = 3\mathbf{i} + 4\mathbf{j}$

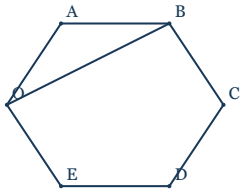
(a) Find the position vector of B.

a

(b) Find the exact value of $|\overrightarrow{OB}|$ in simplified surd form.

b

Exercise 11D - Year 1 Pages 243-244 22



4

OABCDE is a regular hexagon. With O as origin, A and B have position vectors a and b. Find the position vectors of C, D and E.

4(a)

4(b)

4(c)

5

Three vertices of a parallelogram have position vectors

$$\mathbf{u} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} 8 \\ 6 \end{pmatrix}$$

Find all possible position vectors of the fourth vertex.

5

6

The position vectors of A and B are $4\mathbf{i} - 5\mathbf{j}$ and $6\mathbf{i} + 3\mathbf{j}$.

(a) Find $\overrightarrow{AB}$.

6(a)

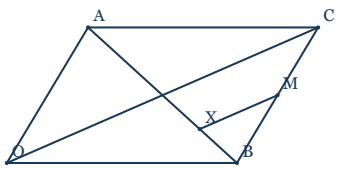
(b) Find $|\overrightarrow{AB}|$.

6(b)

7

A lies on the circle $x^2 + y^2 = 9$. Given $\overrightarrow{OA} = 2k\mathbf{i} + k\mathbf{j}$, find the exact values of k.

7



OACB is a parallelogram with $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$. X lies on AB with AX:XB = 3:1; M is the midpoint of BC.

Show that $\overrightarrow{XM}$ is parallel to $\overrightarrow{OC}$.

Find the two vectors

1

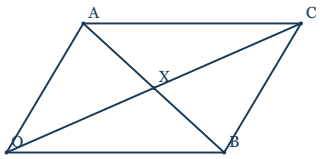
2

Compare the vectors

3

For a proof, find both vectors explicitly and state the scalar-multiple relationship.

Introducing Scalars and Comparing Coefficients 25



OACB is a parallelogram, with $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$. The diagonals OC and AB meet at X. Prove that the diagonals bisect each other.

Express OX by two different routes. Since a and b are nonparallel, their coefficients can be compared.

Along OC

1

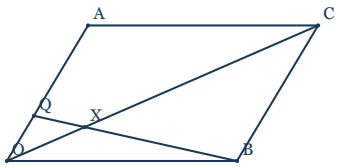
Along OB then BX

2

Compare coefficients

3

Test Your Understanding 26



OACB is a parallelogram with $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$. Q lies on OA with $\overrightarrow{OQ} = \frac{1}{3}\mathbf{a}$. BQ and OC intersect at X.

(a) Let $\overrightarrow{OX} = \lambda\overrightarrow{OC}$. Express OX in terms of a, b and λ.

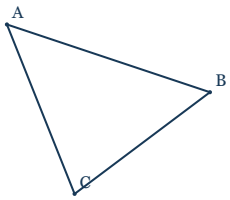
a

(b) Let $\overrightarrow{BX} = \mu\overrightarrow{BQ}$. Express OX in terms of a, b and μ.

b

(c) Find λ and hence the ratio OX:XC.

c



Given $\overrightarrow{AB} = 3\mathbf{i} - 2\mathbf{j}$ and $\overrightarrow{AC} = \mathbf{i} - 5\mathbf{j}$, determine angle BAC.

Find all three side lengths, then use the cosine rule.

Find the lengths

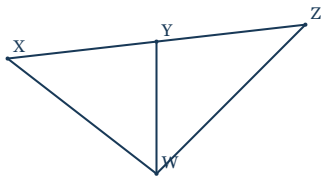
1

Apply the cosine rule

2

Check whether the side lengths identify a special triangle.

Exercise 11E - Year 1 Pages 246-247 28



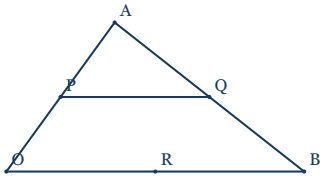
1. Given $\overrightarrow{WX} = \mathbf{a}$, $\overrightarrow{WY} = \mathbf{b}$, $\overrightarrow{WZ} = \mathbf{c}$ and $\overrightarrow{XY} = \overrightarrow{YZ}$, prove that $\mathbf{a} + \mathbf{c} = 2\mathbf{b}$.

Express both equal vectors from the same origin

1

Equate and rearrange

2



2. OAB is a triangle. P, Q and R are the midpoints of OA, AB and OB respectively. Given $\overrightarrow{OP} = \mathbf{p}$ and $\overrightarrow{OR} = \mathbf{r}$:

(a)(i) Find $\overrightarrow{OB}$.

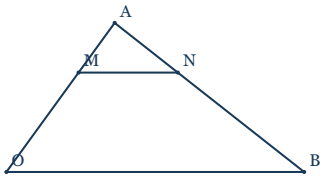
a(i)

(a)(ii) Find $\overrightarrow{PQ}$.

a(ii)

(b) Hence prove that triangle PAQ is similar to triangle OAB.

b



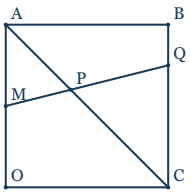
3. OAB is a triangle with $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$. M lies on OA with OM:MA = 2:1. N lies on AB and MN is parallel to OB.

(a) Express ON in terms of a and b.

a

(b) Show that AN:NB = 1:2.

b



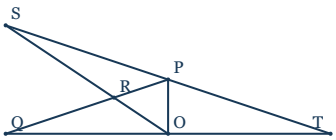
4. OABC is a square. M is the midpoint of OA; Q lies on BC with BQ:QC = 1:3. AC and MQ meet at P. Let $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OC} = \mathbf{c}$.

(a) Express OP in terms of a and c.

a

(b) Show that AP:PC = 2:3.

b



6. OPQ is a triangle. R lies on PQ with $2\overrightarrow{PR} = \overrightarrow{RQ}$, and $\overrightarrow{OS} = 3\overrightarrow{OR}$. Given $\overrightarrow{OP} = \mathbf{a}$ and $\overrightarrow{OQ} = \mathbf{b}$:

(a) Show that OS = 2a + b.

a

(b) T satisfies $\overrightarrow{OT} = -\mathbf{b}$. Prove that T, P and S lie on a straight line.

b

[STEP 2010 Q7] O, A and B are not collinear. Let $\mathbf{c} = \alpha\mathbf{a} + \beta\mathbf{b}$, where $\alpha, \beta > 0$ and $\alpha + \beta < 1$. P $=$ OA $\cap$ BC and Q $=$ OB $\cap$ AC.

Show that $\mathbf{p} = \frac{\alpha\mathbf{a}}{1-\beta}$ and find q. Show that $\mathbf{r} = \frac{\alpha\mathbf{a}+\beta\mathbf{b}}{\alpha+\beta}$ lies on OC and AB. If S $=$ OB $\cap$ PR, prove OQ/BQ $=$ OS/BS.

P and Q

1

R lies on both lines

2

Locate S and compare distances

3

The stated intersection S exists when $\alpha + 2\beta \neq 1$. Otherwise PR is parallel to OB.

Vectors include direction

A velocity of $\begin{pmatrix}3\\4\end{pmatrix}$ km/h means a position change of 3 km east and 4 km north each hour.

Speed

Displacement and distance

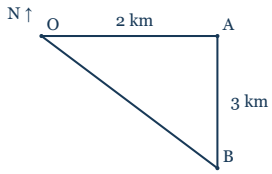
A displacement of $\begin{pmatrix}-5\\12\end{pmatrix}$ km has magnitude

Magnitude

This is the straight-line distance between the endpoints. Distance travelled can be greater if the route is not straight.

Velocity and displacement are vectors. Speed and distance are scalars. Speed is the magnitude of velocity.

Example 35



A girl walks 2 km due east from O to A, then 3 km due south from A to B. Use east and north as the positive component directions.

(a) Find the total distance travelled.

a

(b) Find the position vector of B relative to O.

b

(c) Find $|\overrightarrow{OB}|$.

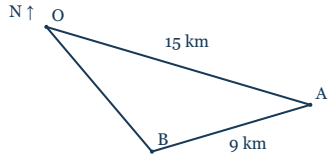
c

(d) Find the bearing of B from O.

d

Bearings are measured clockwise from north. B lies southeast of O, so its bearing must lie between 90° and 180°.

Further Example 36



A cadet walks 15 km from O on a bearing of 120° to A, then 9 km on a bearing of 240° to B. He returns directly to O. Use east and north as positive directions.

(a) Find the position vector of A.

a

For a bearing β and distance d: east component $=$ d $\sin\beta$; north component $=$ d $\cos\beta$.

(b) Find the distance OB.

b

(c) Find the bearing of B from O.

c

Further Example 37

Continuing the cadet example: O to A is 15 km on bearing 120°; A to B is 9 km on bearing 240°.

(d) Find the position vector of B relative to O.

Add the two displacement vectors

d

Keep exact values until the final step. The positive east component and negative north component confirm that B lies southeast of O.

Exercise 11F - Year 1 Pages 250-251 38

5

A particle has mass 0.3 kg and acceleration $\mathbf{a} = (5\mathbf{i} + 7\mathbf{j})\,\text{m s}^{-2}$.

(a) Find the angle between the acceleration and i.

5(a)

(b) Using $\mathbf{F} = m\mathbf{a}$, find the magnitude of the force.

5(b)

6

$\mathbf{F}_1 = (3\mathbf{i} - 4\mathbf{j})\,\text{N}, \quad \mathbf{F}_2 = (p\mathbf{i} + q\mathbf{j})\,\text{N}$

The resultant R is parallel to $2\mathbf{i} - \mathbf{j}$.

(a) Find its angle to i.

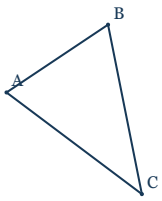
6(a)

(b) Show that $p + 2q = 5$.

6(b)

(c) Given $p = 1$, find the magnitude of R.

6(c)



7. A triangular field ABC has

$\overrightarrow{AB} = (30\mathbf{i} + 40\mathbf{j})$ m, $\overrightarrow{AC} = (40\mathbf{i} - 60\mathbf{j})$ m.

(a) Find BC.

a

(b) Find angle BAC to 1 decimal place.

b

(c) Find the area of the field.

c

Use the unrounded angle in the area calculation.

8. A boat has position vector (2i + j) km and a buoy has position vector (6i − 4j) km. Take i east and j north. The boat travels with constant velocity (8i − 10j) km/h.

(a) Find the distance from boat to buoy.

a

(b) Find the bearing of the boat from the buoy.

b

(c) Verify that the boat is travelling towards the buoy.

c

(d) Find the speed.

d

(e) Find the time to reach the buoy.

e